Annex I. Guinea-Bissau: Debt Sustainability Analysis Update[1]

This update of the joint IMF-World Bank Debt Sustainability Analysis (DSA) indicates that Guinea-Bissau faces a moderate risk of debt distress based on an assessment of public external debt, but a heightened overall risk of debt distress reflecting significant vulnerabilities related to domestic debt. [2,3] *Under the baseline scenario, all external debt present value (PV) ratios show improvements relative to the last DSA, due largely to good revenue performance in 2015 and expected benign developments in the terms of trade and nominal GDP. Under the most extreme shock scenario, the PV of debt-to-exports ratio breaches its threshold for only a short period, unlike under the last DSA where the breach extended over a long period. In contrast, the overall public debt indicator has worsened relative to the last DSA, mainly due to the domestic debt increase on account of a realized contingent liability, new Treasury bill issuance and other new domestic loans—elevating the overall risk of debt distress. The economy remains vulnerable to adverse shocks to exports, which could be amplified by worsening terms of trade or a decline in (or non-realization of projected) foreign direct investment.*

1. Compared to the last DSA, expected improvements in tax administration, public financial management, and the terms of trade are the main differences in the domestic assumptions. Real GDP growth is expected to follow almost exactly the same trajectory as under the last DSA, averaging 5 percent over the projection period (Text Table 1 and Table 1). Tax revenue, which shows a sluggish short-term trajectory, is projected to pick up in the longer term in response to growth in incomes and continuing improvements in tax administration. These expected advances, coupled with parallel improvements in public financial management, are expected to yield gains in the overall fiscal balance, with the primary deficit falling gradually and converting to surpluses by the end of the projection period, 2036.

2. For international prices, the DSA update is based on the IMF WEO medium-term projections of fuel and non-fuel prices, which prop up Guinea-Bissau's terms of trade, export values, and nominal GDP, and help moderate debt vulnerabilities. In particular, as the price of cashew (the main export produce of Guinea-Bissau) is expected to increase into the medium term, the country's terms of trade are projected to strengthen along with its export values. Reflecting the expected benign developments in Guinea-Bissau's terms of trade, export growth is now expected at a higher rate (averaging 10.2 percent of GDP in the long term, compared with 7.5 percent in the last DSA); and import growth is expected to be higher as well, in view of higher expected income and FDI flows. As a result, the external non-interest current account balance is expected to show a lower

[1] This DSA updates the analysis presented to the IMF Executive Board in July 2015. The DSA Update was approved by Messrs. Roger Nord and Peter Allum (IMF) and Ms. Paloma Anos-Casero (IDA).

[2] The staffs of the IMF and IDA, in consultation with the Debt Management Unit of the Bissau-Guinean Ministry of Economy and Finance, prepared this DSA update jointly. The fiscal year of Guinea-Bissau spans January 1–December 31. The previous full DSA was prepared on June 24, 2015 (see IMF Country Report Number 15/194).

[3] Debt sustainability thresholds are determined by the three-year (2013-2015) averages of the Country Policy and Institutional Assessment (CPIA) rating (2.50) that classifies Guinea-Bissau as having weak policy performance and institutional framework.

long-term average deficit than under the last DSA. In parallel, the expected expansion of export values and nominal GDP should help domestic revenue intake, which is now expected to grow at a somewhat faster pace than under the last DSA. As a result of these expectations, and coupled with the Bissau-Guinean authorities' pursuit of grants and highly concessional borrowing, the baseline PV of debt-to-GDP, debt-to-exports and debt-to-revenue ratios are projected to remain below their respective thresholds into the long term, with a similar profile for debt service ratios.

3. The baseline macroeconomic assumptions underlying this DSA update are as follows:

- **Real GDP growth:** Economic growth is expected to remain robust, averaging around 5 percent into the medium term. Over the long term, growth could strengthen with the realization of expected FDI inflows and improvements in growth-enhancing infrastructure, and is projected to be above its historical average rate during the forecast period (Text Table 1).[4]
- **Consumer price inflation** is projected to rise in the medium term, with increasing disposable incomes as the economy sustains its recovery. Following the decline to -1 percent at end 2014, average inflation picked up to 1½ percent at end-2015, and it is now expected to increase to the WAEMU threshold (3 percent) by end-2018 and hover around that figure into the long term.
- **Central government balances:** The primary fiscal deficit is projected to gradually decline in the medium and long term. Specifically, the primary deficit, which increased in 2015 on account of weak public financial management (PFM) in the context of a lingering political crisis, is expected to improve as the authorities strengthen PFM and enhance their tax effort. Thereafter, continued improvements in tax administration and revenue measures should drive a gradual decline in the primary deficit to insignificant levels by 2021, to surpluses by 2026, and yield an average surplus during 2022–36 (Table 3). Non-interest primary expenditure is expected to be contained to average around 19 percent of GDP per year in 2016-21 (this was 20.8 percent in 2015).
- **External current account balance:** Reflecting benign expected terms of trade and increasing export volumes, the current account deficit is projected to remain low, at around 2 percent of GDP in 2016 and to average around 3 percent in 2017–22; it is then expected to remain around this average in the longer term (2023-36), mainly on account of expected overall income growth and increases in imports associated with increasing FDI inflows.
- **Financing flows:** FDI is projected to increase to an average 2.6 percent of GDP per year in 2017–21 (mainly on account of the phosphate project), after hovering below 2.0 percent of GDP in 2015–16. Over the longer term (2022-36), FDI is expected to decline to average 1½ percent of GDP per year (Table 1). This expectation is predicated on improvements in the political situation and the business environment. Official transfers are expected to increase

[4] The authorities have revised real GDP growth rates for 2012–14, based on TA from AFRITAC to finalize the national accounts for these years.

during the projection period, and grant-equivalent financing would form the bulk (some 85–90 percent) of projected external financing.

Text Table 1. Guinea-Bissau: Evolution of Selected Macroeconomic Indicators

	2012	2013	2014	2015	Long Term[1]
Real GDP growth (percent)					
Previous DSA	-1.8	0.8	*2.5*	*4.7*	5.0
Current DSA	-1.7	3.3	2.3	4.8	5.0
Revenue and Grants (percent of GDP)					
Previous DSA	11.4	11.6	21.0	18.7	23.0
Current DSA	11.5	11.3	22.1	20.8	22.6
Primary (non-interest) expenditure (percent of GDP)					
Previous DSA	13.5	13.2	22.0	17.9	24.3
Current DSA	13.6	12.9	23.2	21.6	22.1
Primary balance (percent of GDP)					
Previous DSA	-2.0	-1.6	-1.0	-0.8	-1.3
Current DSA	-2.0	-1.6	-1.1	-0.8	-0.5
Non-interest current account deficit (percent of GDP)					
Previous DSA	8.8	4.3	0.9	3.5	4.6
Current DSA	8.8	7.1	3.2	0.4	2.1
Growth of exports (percent)					
Previous DSA	-44.8	16.3	12.7	13.1	7.5
Current DSA	-44.8	24.7	9.4	41.3	10.2
Growth of imports (percent)					
Previous DSA	-24.4	0.6	24.7	7.9	6.2
Current DSA	-24.4	6.1	15.0	2.5	9.2

Source: Guinea-Bissau authorities and staff estimates.
[1] Long term value of the indicator is defined as an average over the last 15 years of the projections.

4. The DSA indicates an improvement in Guinea-Bissau's external debt sustainability. The downside risks to the macroeconomic outlook under the last DSA, in the form of weakening terms of trade and exports, have not materialized; on the contrary, the country realized an increase in terms of trade and in export volumes, enhancing export values. As a result, all baseline external debt indicators have remained comfortably below their respective policy-dependent indicative thresholds throughout the projection period (2016–36) (Figure 1). Nonetheless, all five external debt indicators are highly sensitive to exports shocks. Under the most extreme shock scenario, one indicator—the PV of debt-to-exports ratio—breached its threshold, albeit by a lower margin and for a shorter period than under the previous DSA. Guinea-Bissau, therefore, remains at moderate risk of debt distress.

5. Due to an increase in domestic debt, however, the overall public debt and debt service indicators have worsened. The government's contracting of additional domestic loans (issuance of Treasury bills totaling CFAF 13 billion) and the realization of a contingent liability (CFAF 9.8 billion)

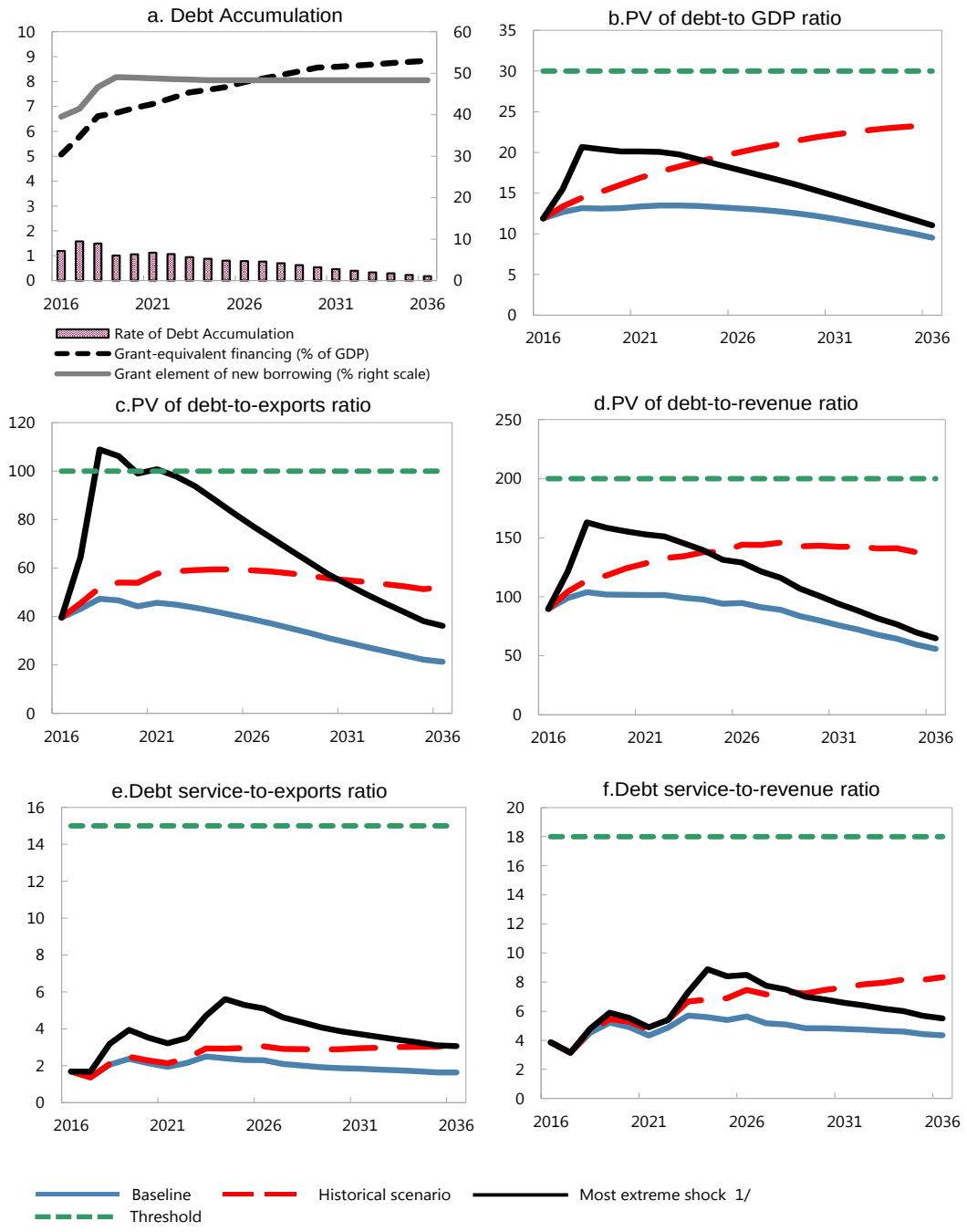
have shifted the baseline PV of public debt-to-GDP ratio upward and its breach of the threshold in the most extreme shock scenario now extends over a longer period than under the last DSA.[5] The new domestic loans have increased public debt by some 3.4 percent of GDP, causing the baseline PV ratio in 2016 to stay above the benchmark for longer than in the previous DSA. The decline in this ratio now occurs at a slower pace than under the last DSA and the fixed primary balance PV ratio also remains above its threshold. In the same vein, the debt service ratios do not decline as fast as under the last DSA, highlighting likely risk of high debt-service burden, particularly in the event of adverse revenue shocks (Figure 2).

6. Results from the updated DSA show that the Bissau-Guinean economy remains vulnerable to adverse shocks. These shocks, including to exports and foreign direct investment (FDI) inflows, and nominal currency depreciation through a strengthening of the US dollar relative to the euro (to which the CFAF is fixed), could result in significant breaches of several thresholds in the short to medium run on a protracted basis. Despite the current benign situation with the external PV of debt-to-exports ratio falling below its threshold after a brief breach and the other ratios being comfortably below their respective ratios, adverse shocks could cause problems of debt distress. Staff therefore reiterates the need for continued prudent borrowing policies, revenue enhancement, sustained improvements in public financial management, and implementation of further growth-enhancing structural reforms.

7. The Bissau-Guinean authorities broadly concur with staff's assessment and policy recommendations. They agree that debt sustainability depends crucially on sound macroeconomic policies that would in turn help their chances of accessing concessional financing. They recognize the contributory role of prudent debt management along with other supportive structural reforms to improve the business environment and to enhance overall growth and export prospects.

[5] In 2015, domestic debt (which was all CFAF denominated) was 58 percent of total public debt excluding contingent liabilities related to the voiding of the bank bailout. It was owed to the BCEAO (46 percent), regional financial institutions (of which banks 42 percent and BOAD the remainder), and the remainder to non-financial institutions (mainly in the form of arrears to suppliers). This configuration is not expected to change significantly over the medium to long term, except for a gradual budgeted clearance of the arrears to domestic suppliers.

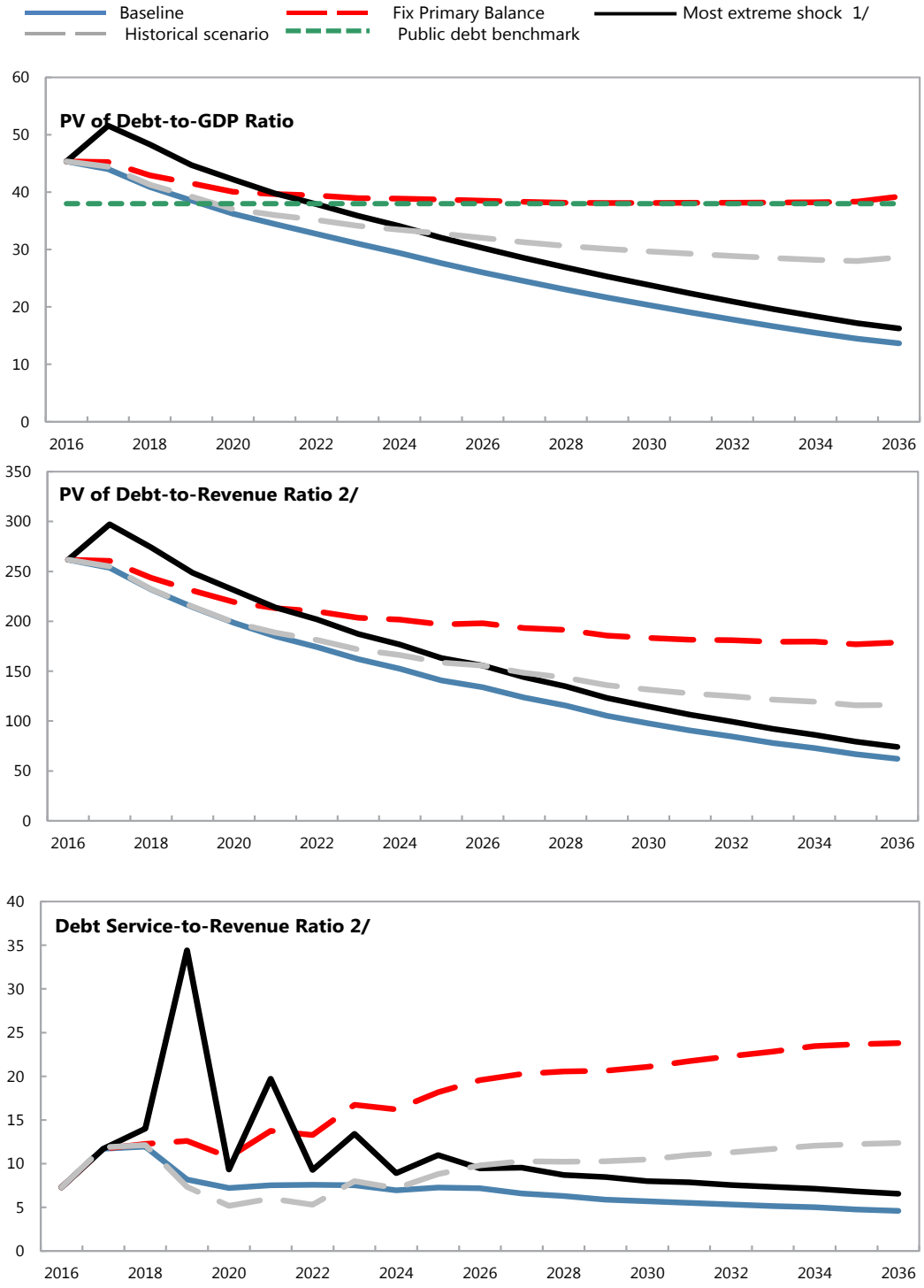
Figure 1. Guinea-Bissau: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2016-2036 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a Exports shock

Figure 2.Guinea-Bissau: Indicators of Public Debt Under Alternative Scenarios, 2016-2036 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2026.
2/ Revenues are defined inclusive of grants.

Table 1. Guinea-Bissau: External Debt Sustainability Framework, Baseline Scenario, 2013–36 [1/]

(percent of GDP unless otherwise indicated)

	Actual			Historical Average [6/]	Standard Deviation [6/]	Projections									
	2013	2014	2015			2016	2017	2018	2019	2020	2021	2016-2021 Average	2026	2036	2022-2036 Average
External debt (nominal) 1/	**16.2**	**14.9**	**15.0**			**14.1**	**14.5**	**15.4**	**15.9**	**16.3**	**16.9**		**18.1**	**13.9**	
of which: public and publicly guaranteed (PPG)	16.1	14.9	14.9			14.1	14.5	15.4	15.9	16.3	16.9		18.1	13.9	
Change in external debt	-10.6	-1.2	0.0			-0.9	0.4	0.9	0.5	0.5	0.6		0.1	-0.7	
Identified net debt-creating flows	3.7	1.4	-0.5			0.3	0.5	1.3	0.9	0.9	0.9		0.4	-0.4	
Non-interest current account deficit	**7.1**	**3.2**	**0.4**	**4.5**	**3.6**	**2.6**	**4.0**	**4.5**	**3.8**	**3.7**	**3.7**		**2.7**	**1.0**	2.1
Deficit in balance of goods and services	10.5	13.0	5.9			5.8	7.5	7.9	7.1	6.5	6.3		5.9	2.9	
Exports	18.1	19.8	29.0			30.0	29.4	27.8	28.1	29.8	29.3		33.8	44.8	
Imports	28.6	32.8	34.9			35.9	36.8	35.7	35.3	36.3	35.6		39.7	47.7	
Net current transfers (negative = inflow)	-3.6	-7.4	-3.6	-6.4	2.5	-1.1	-1.6	-1.5	-1.5	-1.2	-1.3		-1.9	-0.9	-1.5
of which: official	-0.8	-4.5	-0.9			-0.2	-0.5	-0.5	-0.5	-0.5	-0.5		-0.5	-0.5	
Other current account flows (negative = net inflow)	0.3	-2.5	-1.9			-2.1	-1.9	-2.0	-1.9	-1.6	-1.4		-1.3	-1.0	
Net FDI (negative = inflow)	**-1.9**	**-2.0**	**-1.8**	**-2.0**	**0.8**	**-2.0**	**-2.9**	**-2.9**	**-2.5**	**-2.4**	**-2.3**		**-1.7**	**-0.9**	-1.5
Endogenous debt dynamics 2/	**-1.5**	**0.3**	**0.8**			**-0.3**	**-0.6**	**-0.3**	**-0.4**	**-0.5**	**-0.5**		**-0.6**	**-0.5**	
Contribution from nominal interest rate	0.1	0.3	0.3			0.3	0.1	0.3	0.4	0.3	0.2		0.3	0.2	
Contribution from real GDP growth	-0.8	-0.4	-0.7			-0.6	-0.7	-0.7	-0.7	-0.7	-0.8		-0.8	-0.7	
Contribution from price and exchange rate changes	-0.8	0.3	1.3			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-14.3**	**-2.6**	**0.6**			**-1.1**	**-0.2**	**-0.3**	**-0.5**	**-0.4**	**-0.3**		**-0.4**	**-0.3**	
of which: exceptional financing	0.0	-0.6	0.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	12.3			11.9	12.7	13.2	13.1	13.2	13.4		13.1	9.5	
In percent of exports	...	...	42.3			39.6	43.2	47.3	46.6	44.2	45.6		38.8	21.3	
PV of PPG external debt	**...**	**...**	**12.3**			**11.9**	**12.7**	**13.2**	**13.1**	**13.2**	**13.4**		**13.1**	**9.5**	
In percent of exports	**...**	**...**	**42.2**			**39.6**	**43.1**	**47.3**	**46.6**	**44.2**	**45.6**		**38.8**	**21.3**	
In percent of government revenues	**...**	**...**	**87.0**			**89.7**	**98.9**	**103.9**	**101.9**	**101.6**	**101.5**		**94.7**	**56.0**	
Debt service-to-exports ratio (in percent)	**1.8**	**2.3**	**1.5**			**1.7**	**1.4**	**2.1**	**2.4**	**2.1**	**1.9**		**2.3**	**1.6**	
PPG debt service-to-exports ratio (in percent)	**1.8**	**2.3**	**1.5**			**1.7**	**1.4**	**2.1**	**2.4**	**2.1**	**1.9**		**2.3**	**1.6**	
PPG debt service-to-revenue ratio (in percent)	**4.1**	**3.6**	**3.1**			**3.8**	**3.1**	**4.5**	**5.2**	**4.9**	**4.3**		**5.6**	**4.3**	
Total gross financing need (Millions of U.S. dollars)	58.8	16.8	-9.3			12.3	18.6	28.3	27.4	29.6	31.8		39.6	38.4	
Non-interest current account deficit that stabilizes debt ratio	17.7	4.4	0.4			3.5	3.6	3.5	3.3	3.2	3.1		2.6	1.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.3	2.3	4.8	3.4	2.4	4.8	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
GDP deflator in US dollar terms (change in percent)	3.0	-1.9	-8.1	2.8	11.3	6.3	1.0	2.2	2.9	3.0	2.0	2.9	2.0	2.0	2.0
Effective interest rate (percent) 5/	0.3	1.9	1.7	0.8	0.7	2.5	0.4	2.4	2.5	1.8	1.5	1.9	1.7	1.4	1.6
Growth of exports of G&S (US dollar terms, in percent)	24.7	9.4	41.3	16.0	31.2	15.2	3.6	1.7	9.2	14.5	5.4	8.3	10.6	5.9	10.2
Growth of imports of G&S (US dollar terms, in percent)	6.1	15.0	2.5	9.8	14.8	14.4	8.9	4.1	6.6	11.3	5.1	8.4	9.5	6.1	9.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	39.5	41.5	46.7	49.1	48.9	48.8	45.7	48.3	48.3	48.4
Government revenues (excluding grants, in percent of GDP)	7.9	12.6	14.1			13.2	12.8	12.7	12.9	12.9	13.2		13.9	17.0	15.0
Aid flows (in Millions of US dollars) 7/	45.2	126.7	84.9			57.9	71.9	96.1	107.8	119.6	130.9		201.9	419.0	
of which: Grants	35.9	100.3	67.9			52.4	61.8	72.0	80.4	90.0	99.1		162.4	380.6	
of which: Concessional loans	9.2	26.4	17.0			5.4	10.1	24.1	27.4	29.6	31.8		39.6	38.4	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.1	5.8	6.6	6.7	6.9	7.1		8.0	8.8	8.3
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			88.3	86.5	85.0	87.1	87.4	87.6		89.9	95.3	91.8
Memorandum items:															
Nominal GDP (Millions of US dollars)	1052.5	1055.9	1017.3			1133.1	1201.4	1289.5	1392.7	1506.1	1613.8		2274.0	4515.3	
Nominal dollar GDP growth	6.3	0.3	-3.7			11.4	6.0	7.3	8.0	8.1	7.2	8.0	7.1	7.1	7.1
PV of PPG external debt (in Millions of US dollars)			122.3			134.4	152.3	170.2	183.3	198.0	215.1		297.8	429.3	
(PVt-PVt-1)/GDPt-1 (in percent)						1.2	1.6	1.5	1.0	1.1	1.1	1.2	0.8	0.2	0.6
Gross workers' remittances (Millions of US dollars)	29.9	31.0	26.7			27.6	28.5	29.6	30.7	31.8	33.0		39.7	57.3	
PV of PPG external debt (in percent of GDP + remittances)	...	...	12.0			11.6	12.4	12.9	12.8	12.9	13.1		12.9	9.4	
PV of PPG external debt (in percent of exports + remittances)	...	...	38.7			36.6	39.9	43.7	43.2	41.2	42.7		36.9	20.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.4			1.6	1.3	1.9	2.2	2.0	1.8		2.2	1.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Guinea-Bissau: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016–36

(percent)

	Projections							
	2016	2017	2018	2019	2020	2021	**2026**	2036
PV of debt-to GDP ratio								
Baseline	12	13	13	13	13	13	**13**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	12	13	14	15	16	17	**20**	23
A2. New public sector loans on less favorable terms in 2016-2036 2	12	13	14	14	15	15	**17**	14
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	12	13	14	14	14	14	**14**	10
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	12	16	21	20	20	20	**18**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	12	14	16	16	16	16	**16**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	12	12	12	12	12	13	**13**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	12	13	15	15	15	15	**15**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	12	18	19	19	19	19	**19**	14
PV of debt-to-exports ratio								
Baseline	40	43	47	47	44	46	**39**	21
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	40	45	52	54	54	58	**59**	52
A2. New public sector loans on less favorable terms in 2016-2036 2	40	44	50	51	49	52	**50**	32
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	40	43	47	47	44	46	**39**	21
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	40	65	109	106	99	101	**77**	36
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	40	43	47	47	44	46	**39**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	40	42	45	44	42	43	**37**	21
B5. Combination of B1-B4 using one-half standard deviation shocks	40	43	47	47	44	46	**39**	22
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	40	43	47	47	44	46	**39**	21
PV of debt-to-revenue ratio								
Baseline	90	99	104	102	102	101	**95**	56
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	90	104	114	118	124	128	**144**	137
A2. New public sector loans on less favorable terms in 2016-2036 2	90	101	110	111	114	116	**122**	84
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	90	103	113	111	110	110	**102**	60
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	90	121	163	158	155	153	**129**	65
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	90	109	128	126	125	125	**116**	69
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	90	96	98	97	96	96	**91**	55
B5. Combination of B1-B4 using one-half standard deviation shocks	90	102	115	113	112	112	**106**	64
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	90	141	148	146	145	144	**134**	79

Table 2. Guinea-Bissau: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016–36 (continued)

(percent)

Debt service-to-exports ratio								
Baseline	2	1	2	2	2	2	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	2	1	2	2	2	2	**3**	3
A2. New public sector loans on less favorable terms in 2016-2036 2	2	1	2	3	2	2	**3**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	2	1	2	2	2	2	**2**	2
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	2	2	3	4	4	3	**5**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	2	1	2	2	2	2	**2**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	2	1	2	2	2	2	**2**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	2	1	2	2	2	2	**2**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	2	1	2	2	2	2	**2**	2
Debt service-to-revenue ratio								
Baseline	4	3	5	5	5	4	**6**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	4	3	5	5	5	5	**7**	8
A2. New public sector loans on less favorable terms in 2016-2036 2	4	3	5	6	6	5	**7**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	4	3	5	6	5	5	**6**	5
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	4	3	5	6	6	5	**8**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	4	3	6	6	6	5	**7**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	4	3	4	5	5	4	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	4	3	5	6	6	5	**6**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	4	4	6	7	7	6	**8**	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	47	47	47	47	47	47	**47**	47

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Guinea-Bissau: Public Sector Debt Sustainability Framework, Baseline Scenario, 2013–36

(percent of GDP unless otherwise indicated)

	Actual					Estimate	Projections								
	2013	2014	2015	Average 5/	Standard Deviation 5/	2016	2017	2018	2019	2020	2021	2016-21 Average	2026	2036	2022-36 Average
Public sector debt 1/	48.9	50.5	48.5			49.0	47.2	44.5	42.6	40.6	39.2		32.0	18.8	
of which: foreign-currency denominated	16.1	14.9	14.9			14.1	14.5	15.4	15.9	16.3	16.9		18.1	13.9	
Change in public sector debt	-7.5	1.7	-2.0			0.5	-1.8	-2.7	-1.9	-2.0	-1.5		-1.5	-1.2	
Identified debt-creating flows	-1.4	2.7	-4.1			-2.4	-1.8	-1.4	-1.9	-1.6	-2.4		-2.3	-2.4	
Primary deficit	1.6	1.1	0.8	0.5	3.0	2.5	0.8	1.3	0.9	1.1	0.1	1.1	-0.4	-1.3	-0.5
Revenue and grants	11.3	22.1	20.8			17.9	17.9	18.3	18.6	18.9	19.3		21.0	25.5	
of which: grants	3.4	9.5	6.7			4.6	5.1	5.6	5.8	6.0	6.1		7.1	8.4	
Primary (noninterest) expenditure	12.9	23.2	21.6			20.3	18.8	19.6	19.6	20.0	19.4		20.7	24.2	
Automatic debt dynamics	-2.5	2.1	-4.5			-4.4	-2.3	-2.4	-2.4	-2.4	-2.2		-1.8	-1.1	
Contribution from interest rate/growth differential	-1.9	-0.3	-5.1			-3.6	-2.4	-2.4	-2.3	-2.3	-2.2		-1.8	-1.1	
of which: contribution from average real interest rate	-0.1	0.8	-2.7			-1.3	-0.1	-0.1	-0.2	-0.3	-0.3		-0.2	-0.2	
of which: contribution from real GDP growth	-1.8	-1.1	-2.3			-2.2	-2.3	-2.2	-2.1	-2.0	-1.9		-1.6	-0.9	
Contribution from real exchange rate depreciation	-0.6	2.4	0.6			...	...	...	...	...	...		...	...	
Other identified debt-creating flows	-0.4	-0.4	-0.5			-0.4	-0.4	-0.4	-0.3	-0.3	-0.3		-0.2	-0.1	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.5	-0.5	-0.5			-0.5	-0.4	-0.4	-0.4	-0.3	-0.3		-0.2	-0.1	
Other (specify, e.g. bank recapitalization)	0.0	0.1	0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-6.1	-1.1	2.1			2.8	0.1	-1.3	0.0	-0.4	0.9		0.9	1.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	45.8			46.8	45.4	42.2	39.8	37.5	35.6		27.0	14.4	
of which: foreign-currency denominated	...	...	12.3			11.9	12.7	13.2	13.1	13.2	13.4		13.1	9.5	
of which: external	...	...	12.3			11.9	12.7	13.2	13.1	13.2	13.4		13.1	9.5	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	8.1	7.6	7.0			8.4	6.8	7.1	5.8	5.6	4.3		2.3	-0.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	220.6			261.7	253.0	231.4	213.7	198.0	184.4		128.5	56.7	
PV of public sector debt-to-revenue ratio (in percent)	...	...	325.0			353.1	354.8	333.4	309.7	289.3	270.4		194.7	84.7	
of which: external 3/	...	...	87.0			89.7	98.9	103.9	101.9	101.6	101.5		94.7	56.0	
Debt service-to-revenue and grants ratio (in percent) 4/	6.1	2.6	5.1			7.3	11.7	11.9	8.2	7.2	7.5		6.9	4.1	
Debt service-to-revenue ratio (in percent) 4/	8.8	4.6	7.6			9.8	16.4	17.2	11.8	10.5	11.0		10.4	6.1	
Primary deficit that stabilizes the debt-to-GDP ratio	9.1	-0.5	2.8			2.0	2.6	4.0	2.8	3.0	1.6		1.1	-0.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.3	2.3	4.8	3.4	2.4	4.8	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Average nominal interest rate on forex debt (in percent)	0.3	1.9	1.7	0.8	0.7	2.5	0.4	2.4	2.5	1.8	1.5	1.9	1.7	1.4	1.6
Average real interest rate on domestic debt (in percent)	0.8	2.2	-8.3	-2.8	5.7	-4.6	0.5	-0.5	-0.8	-0.9	-0.8	-1.2	-1.0	-1.2	-1.1
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.5	15.0	4.2	-0.5	8.2	-6.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-0.3	-2.0	10.1	3.6	6.4	6.0	0.9	2.0	2.2	2.2	2.0	2.6	2.0	2.0	2.0
Growth of real primary spending (deflated by GDP deflator, in percent)	-1.9	84.3	-2.4	8.0	26.8	-1.3	-3.1	9.4	4.9	7.4	1.9	3.2	5.5	3.8	6.6
Grant element of new external borrowing (in percent)	...	...	...	...	...	39.5	41.5	46.7	49.1	48.9	48.8	45.7	48.3	48.3	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Guinea-Bissau: Sensitivity Analysis for Key Indicators of Public Debt, 2016–36

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
PV of Debt-to-GDP Ratio								
Baseline	47	45	42	40	37	36	27	14
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	47	46	43	40	38	37	33	29
A2. Primary balance is unchanged from 2016	47	47	44	43	41	41	40	40
A3. Permanently lower GDP growth 1/	47	46	43	41	39	37	31	25
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	47	48	47	45	43	42	35	27
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	47	47	46	43	40	38	29	16
B3. Combination of B1-B2 using one half standard deviation shocks	47	48	46	44	42	40	33	23
B4. One-time 30 percent real depreciation in 2017	47	50	47	44	41	39	28	15
B5. 10 percent of GDP increase in other debt-creating flows in 2017	47	53	50	46	43	41	31	17
PV of Debt-to-Revenue Ratio 2/								
Baseline	262	253	231	214	198	184	128	57
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	262	254	231	214	198	188	148	101
A2. Primary balance is unchanged from 2016	262	260	243	230	219	212	189	156
A3. Permanently lower GDP growth 1/	262	254	234	218	203	191	144	96
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	262	262	251	236	222	211	163	102
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	262	264	251	231	213	198	138	62
B3. Combination of B1-B2 using one half standard deviation shocks	262	263	247	231	217	204	154	89
B4. One-time 30 percent real depreciation in 2017	262	280	256	234	216	200	134	57
B5. 10 percent of GDP increase in other debt-creating flows in 2017	262	295	272	247	230	212	149	67
Debt Service-to-Revenue Ratio 2/								
Baseline	7	12	12	8	7	7	7	4
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	7	12	12	7	5	6	9	10
A2. Primary balance is unchanged from 2016	7	12	12	13	11	14	18	21
A3. Permanently lower GDP growth 1/	7	12	12	8	8	8	9	11
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	7	12	13	10	10	11	13	11
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	7	12	12	15	13	11	8	5
B3. Combination of B1-B2 using one half standard deviation shocks	7	12	13	11	9	11	11	9
B4. One-time 30 percent real depreciation in 2017	7	12	13	10	9	10	10	7
B5. 10 percent of GDP increase in other debt-creating flows in 2017	7	12	14	34	9	19	9	6

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.